

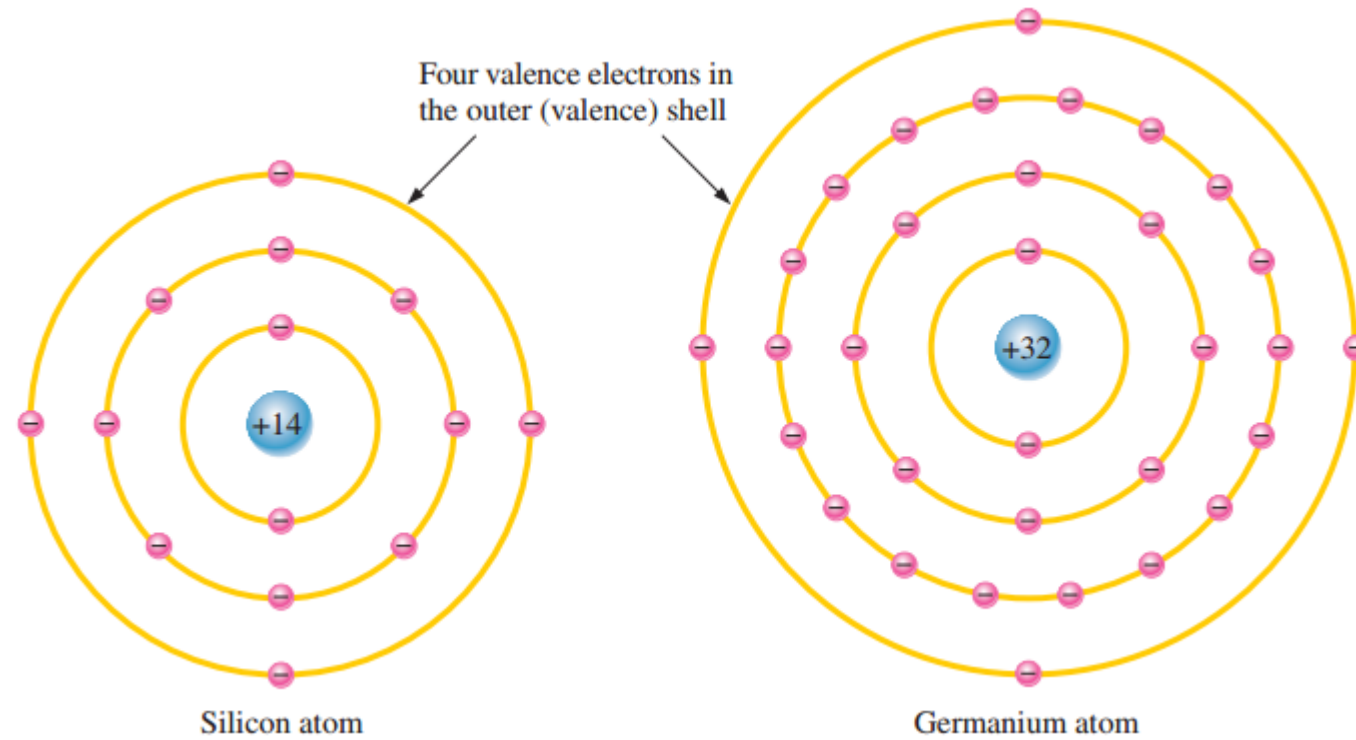
Semiconductors

- What is conductor ?
- Copper (29 e-) is a good conductor as it has only one electron in its valence band
- Similarly the materials which have 4 electrons in their valence band are semiconductor materials
- Examples are Si (14 e-), Germanium (32 e-), Carbon (4 e-) etc.

Semiconductors (contd.)

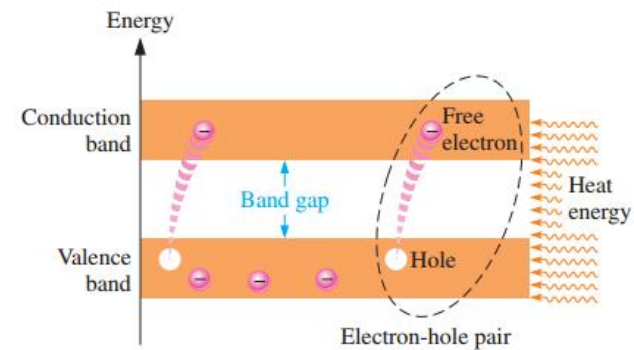
- Why silicon is widely used ?
- Reason is the atomic structure of both materials
- Silicon has 4 electrons in its 3rd shell while Germanium also has 4 electrons but in 4th shell
- Germanium valence e- require small energy to escape from the atom
- This makes Germanium unstable at high temperatures
- That's why Silicon is most widely used in electronics

Semiconductors (contd.)

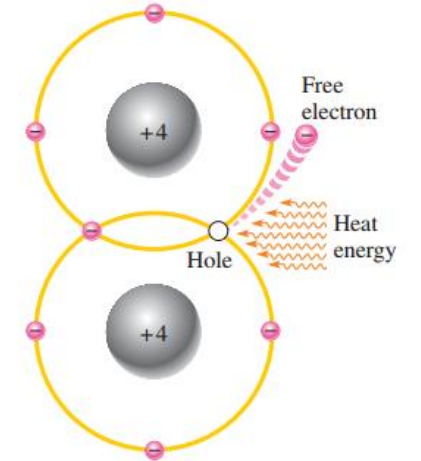


Semiconductors (contd.)

- Concept of *Hole*
- At room temp, some valence electrons absorb energy and jump to the conduction band
- This causes a vacancy in the valence band of crystal called *hole*
- **Recombination** is the process when an e^- falls in the hole



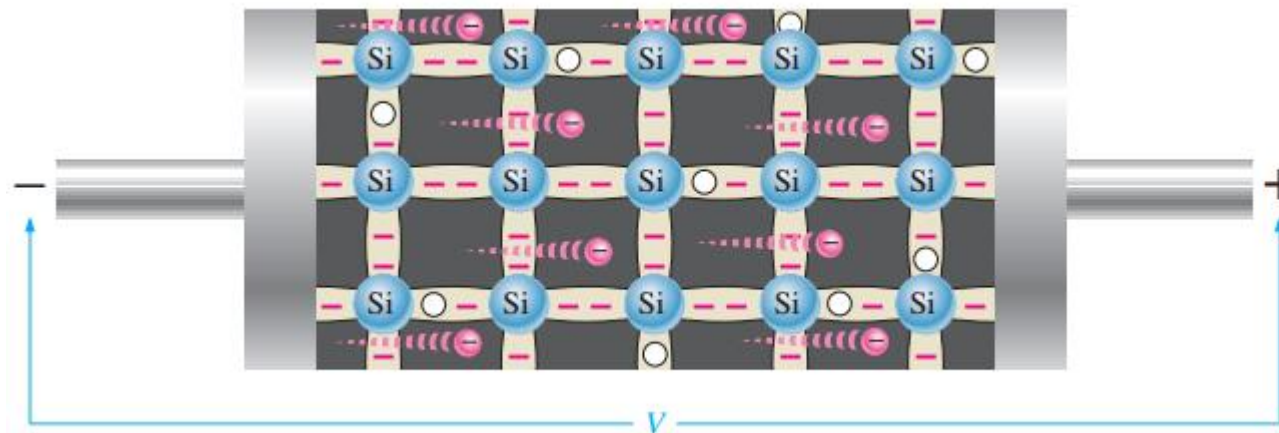
(a) Energy diagram



(b) Bonding diagram

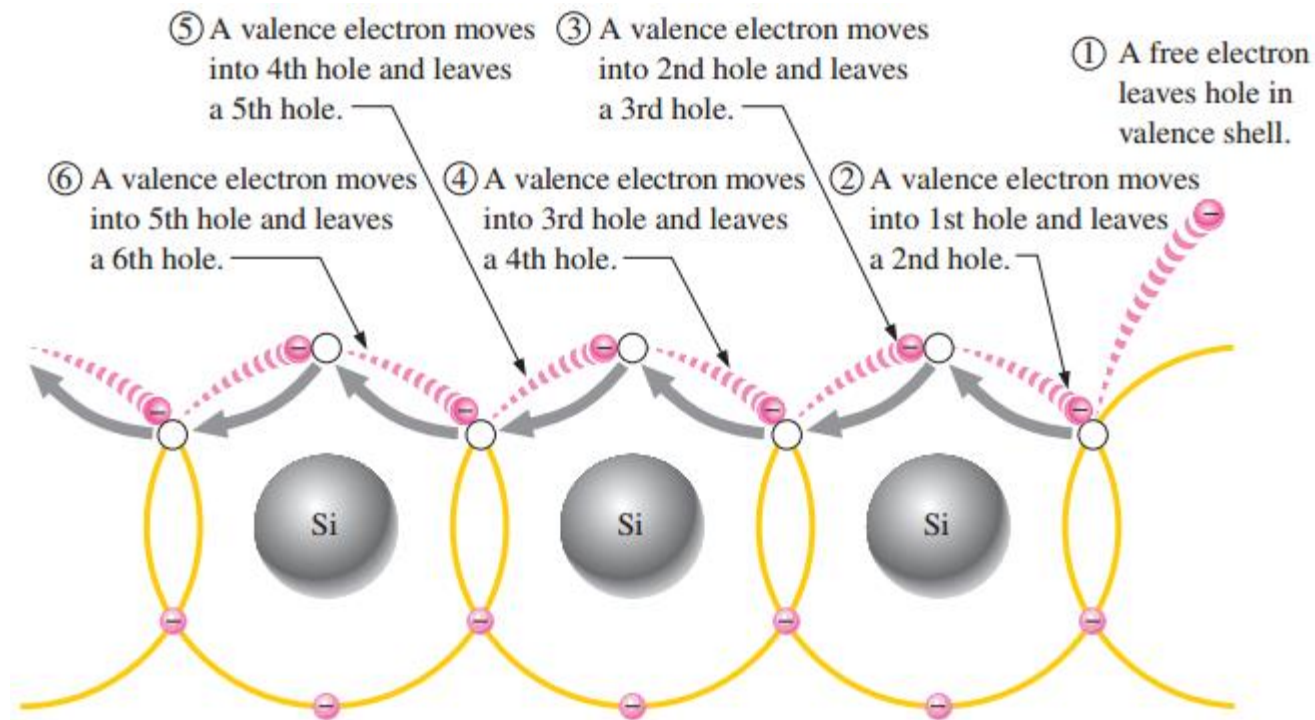
Semiconductors (contd.)

- When voltage is applied to pure semiconductor, then e⁻ can easily move towards positive side
- Known as **electron current**



Semiconductors (contd.)

- Other type is *hole current*, which is explained well in diagram

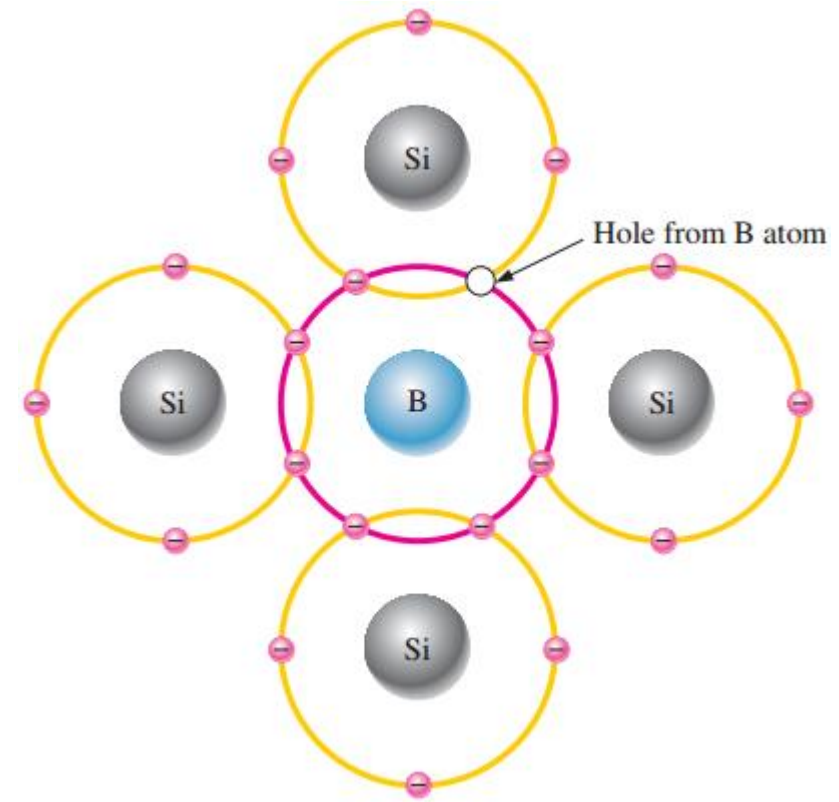
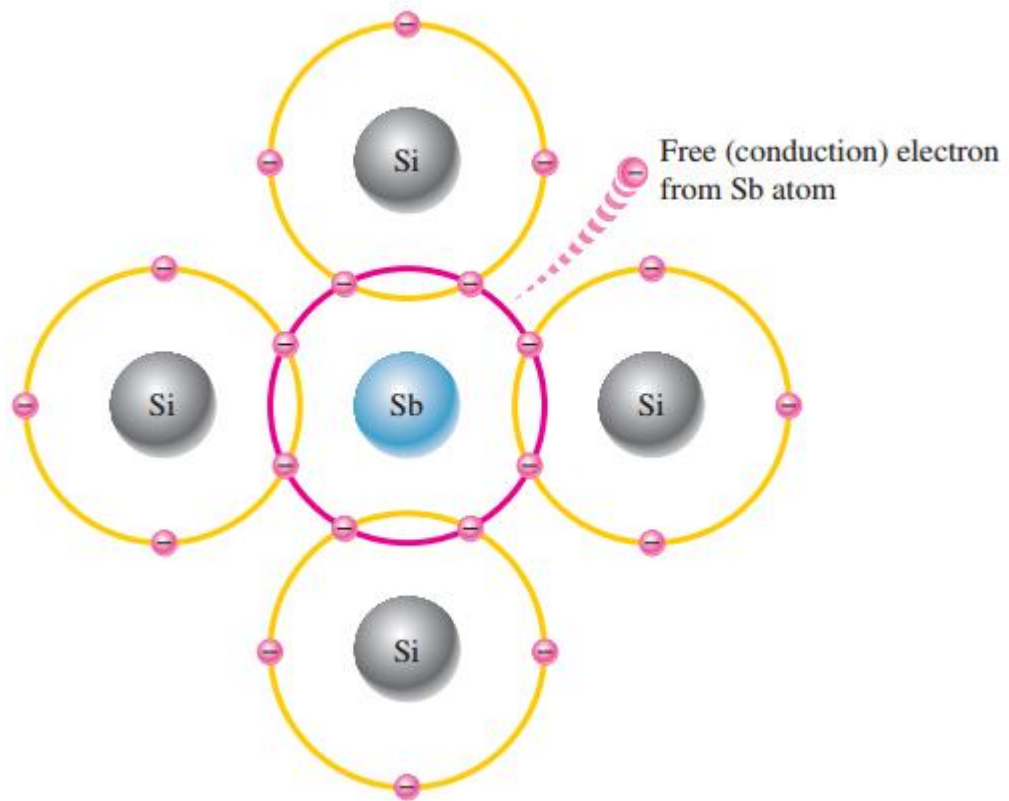


When a valence electron moves left to right to fill a hole while leaving another hole behind, the hole has effectively moved from right to left. Gray arrows indicate effective movement of a hole.

Semiconductors (contd.)

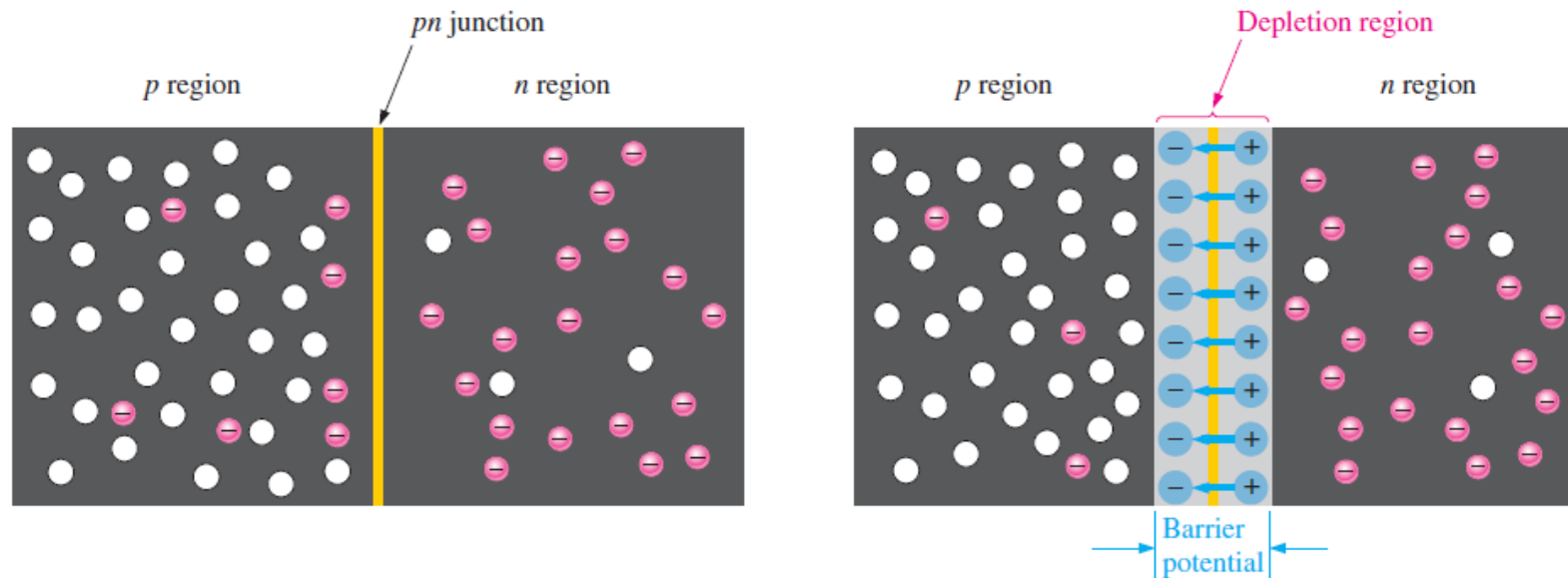
- There are **two types** of semiconductors
 - **n-type semiconductor**: in which **pentavalent materials** added by **doping** to achieve certain electrical characteristics
 - Doped materials can be: **Arsenic, Phosphorous, Bismuth and Antimony** and called **donner atoms**
- **p-type semiconductor**: in which **trivalent materials** added by **doping** to achieve certain electrical characteristics
- Doped materials can be: **Boron, Indium and Gallium** and called acceptor atoms

Semiconductors (contd.)



Diode Theory

- Intrinsic semiconductor doped with trivalent and pentavalent material, a boundary called **pn-junction** is formed between the **p-type** and **n-type** material
- **Diode** created.....

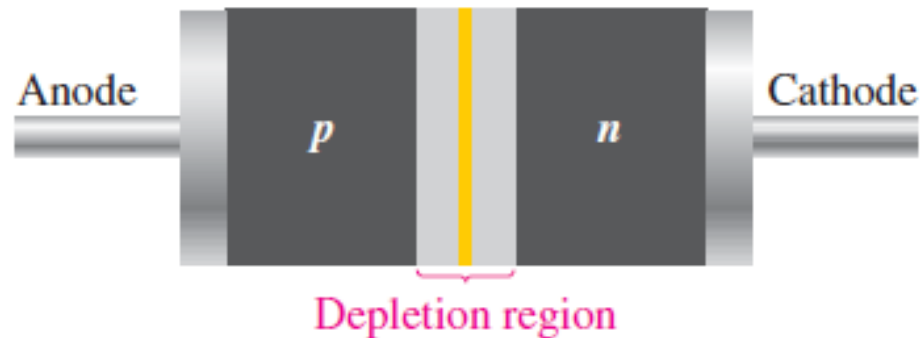


Diode Theory (contd.)

- For every electron which diffuses at the boundary, a **positive charge** is left in the **n-region** and a **negative charge** is left in the **p-region**
- This is **barrier potential** of diode which forbids further diffusion
- The region where this electron hole recombination occurs is called **depletion region**
- Certain amount of voltage equal to barrier potential is required to flow the electrons across the junction
- Typical barrier potential for **Silicon** diode is **0.7V** and for **Germanium** **0.3V** at **25°C**

Diode Theory (contd.)

- Typical diode structure and symbol is shown in fig
- **p-type** region is called **Anode** and **n-type** region is called **Cathode**
- pn-junction is in between the Anode and Cathode



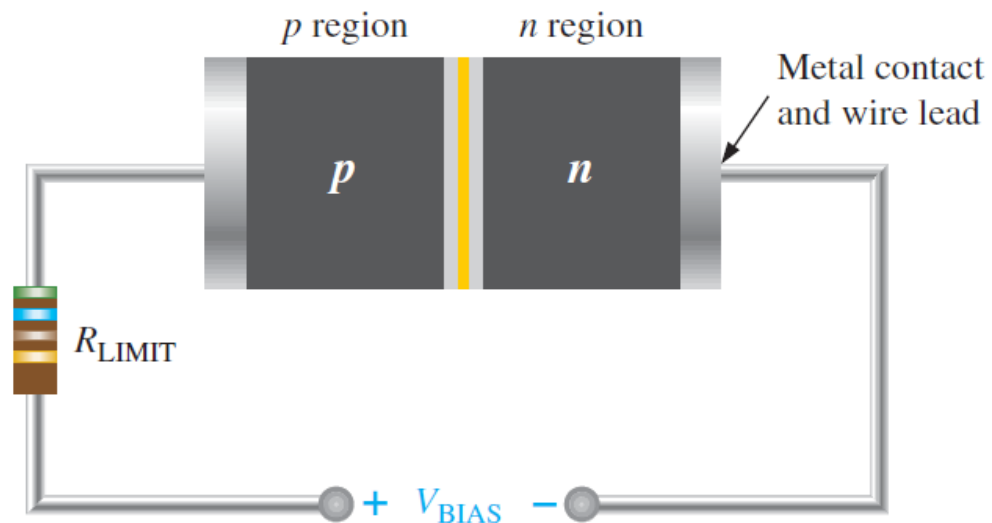
(a) Basic structure



(b) Symbol

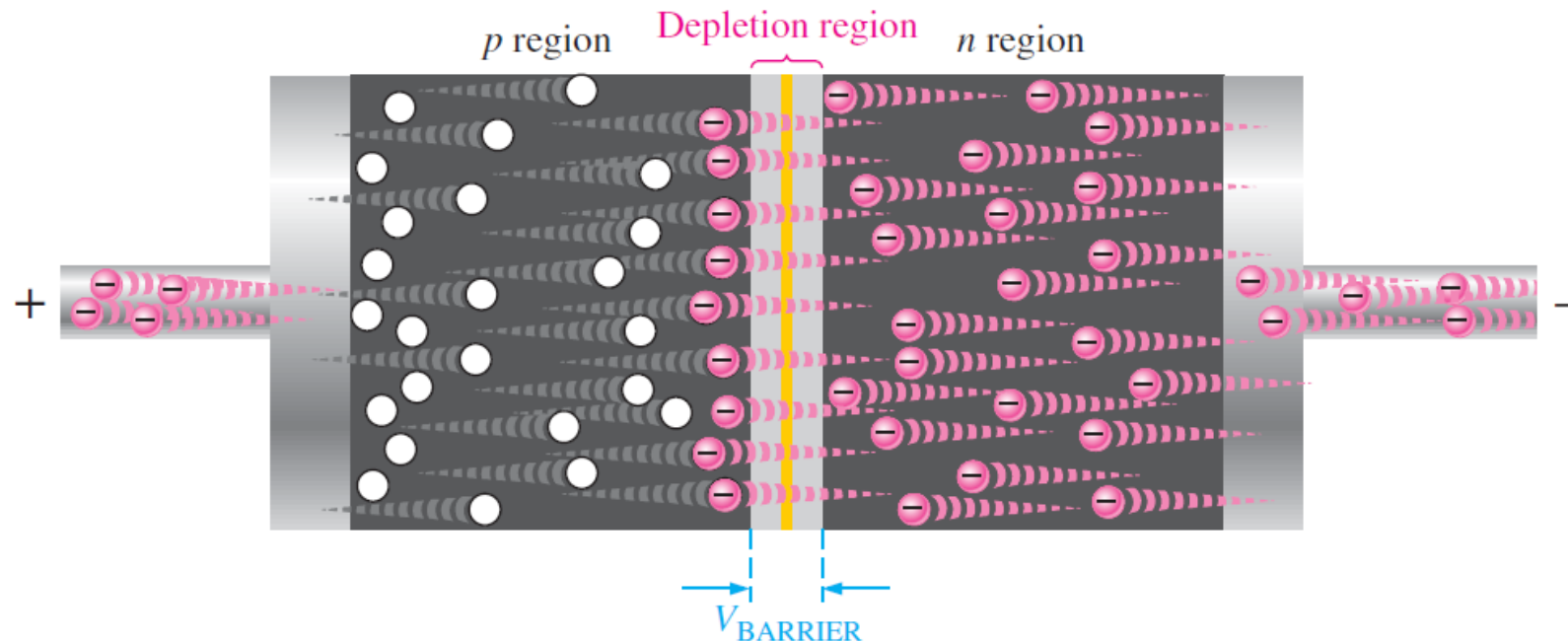
Diode Theory (contd.)

- Biasing of diode is when it is connected with a voltage source
- When n-type material is connected with -ive and p-type material is connected with +ive side of source, it is called forward biasing
- V_{bias} should be greater than the barrier potential



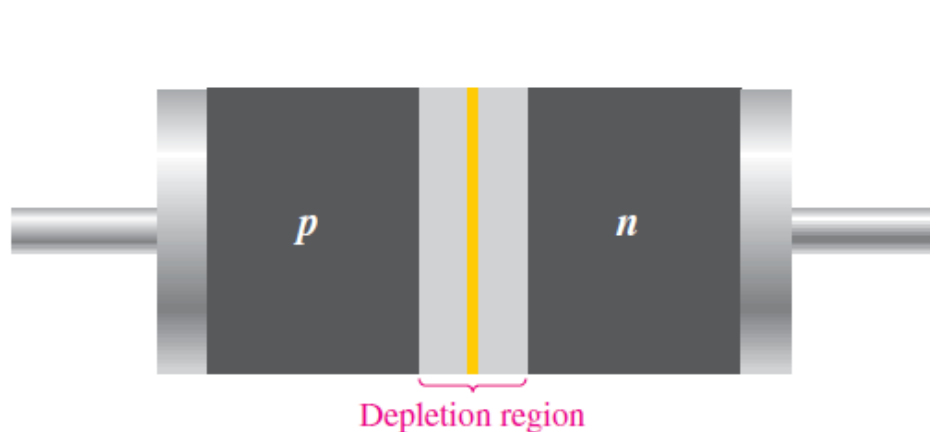
Diode Theory (contd.)

- when voltage is greater than the barrier potential, free electrons cross the barrier potential and move into the p-type material
- Electron current induced inside the diode

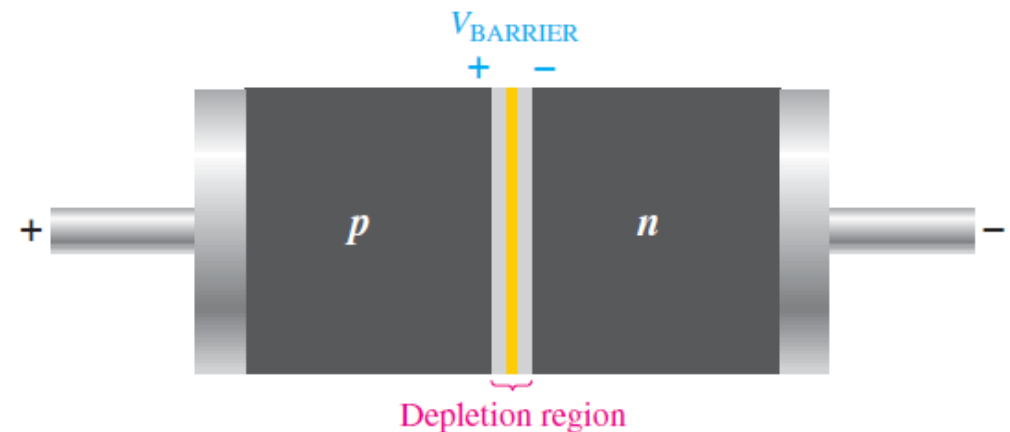


Diode Theory (contd.)

- More electron flow towards the depletion region, positive charge reduce and same is true for holes
- This causes the depletion region to becomes narrow
- Also the concept of energy hill



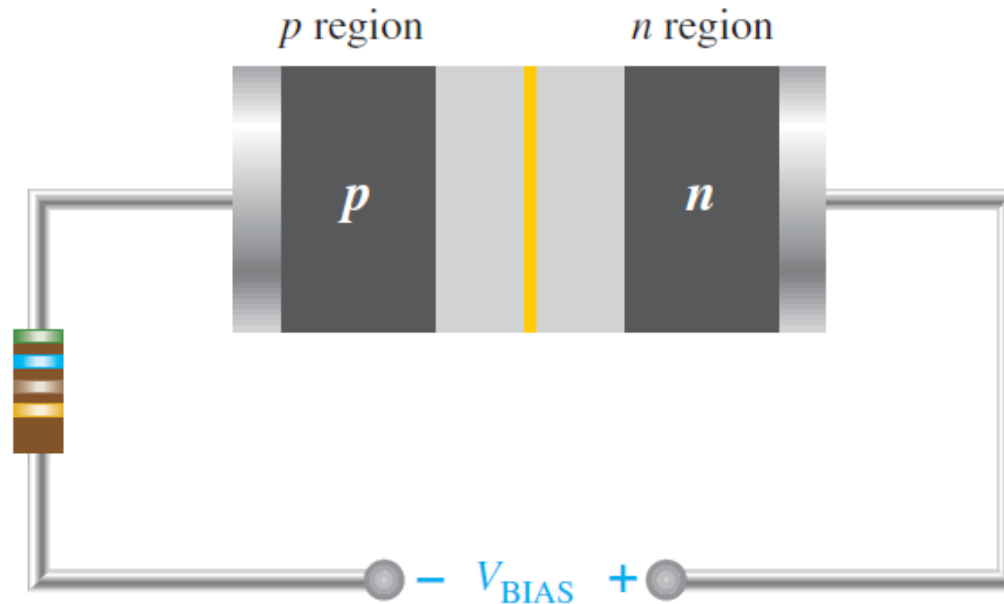
(a) At equilibrium (no bias)



(b) Forward bias narrows the depletion region and produces a voltage drop across the pn junction equal to the barrier potential.

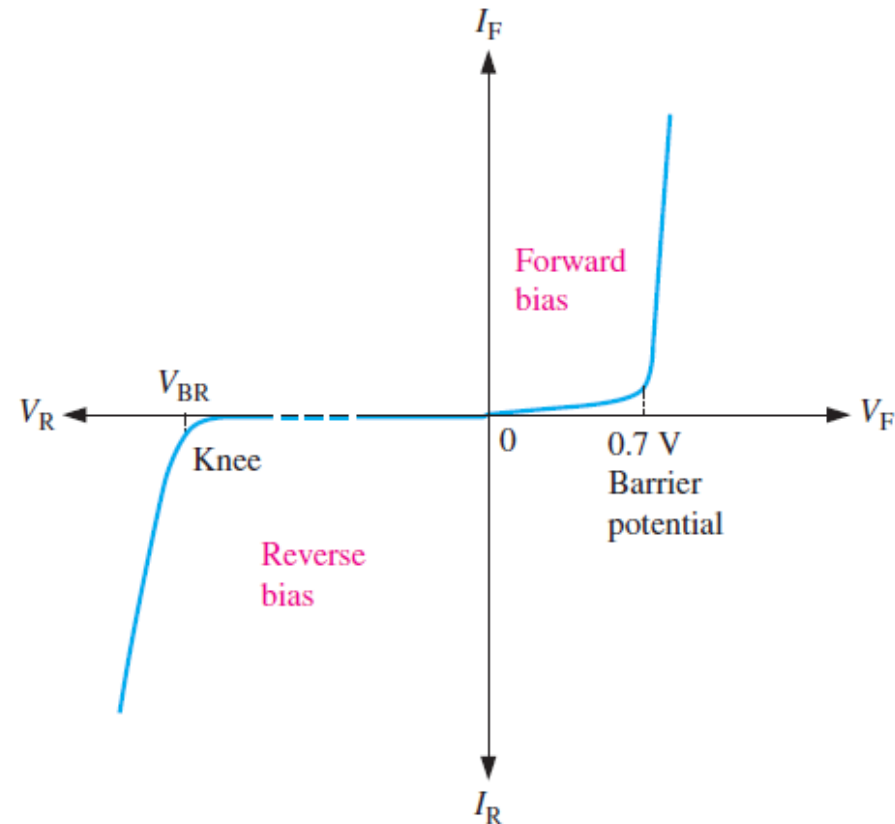
Diode Theory (contd.)

- Reverse biasing is the condition prevents the flow of current through diode
- When p-type is connected with -ive and n-type is connected with +ive end of source



Diode Theory (contd.)

- V-I curve of diode is shown in figure

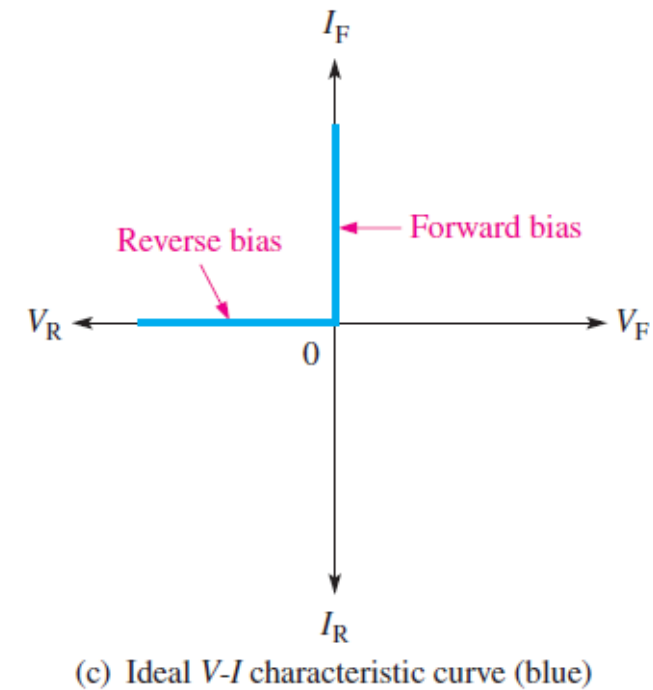
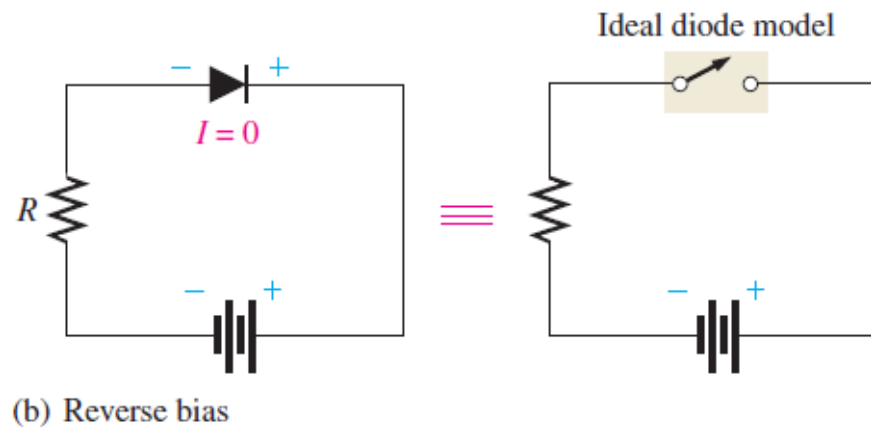
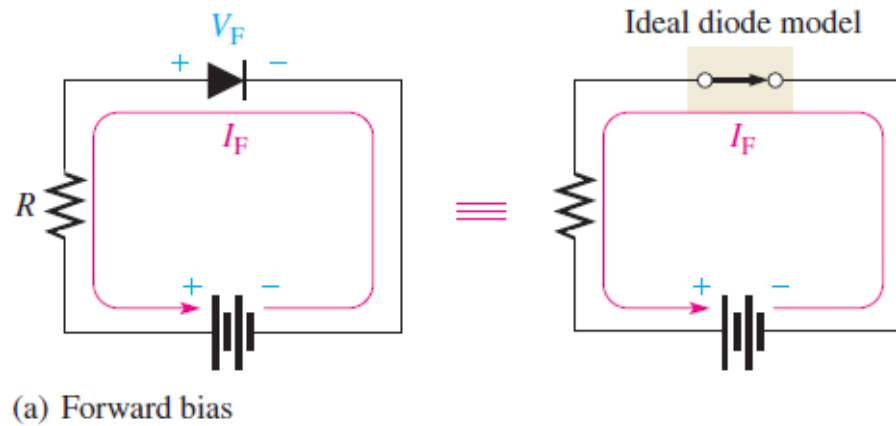


Diode Models

- Ideal Diode Model

- It is the least accurate approximate model
- The diode can be replaced by a simple switch
- When diode is forward bias, diode acts like a closed switch
- When diode is reverse biased, diode acts like an open switch
- The barrier potential, dynamic resistance of diode and reverse current are neglected
- Only used for troubleshooting purpose, whether diode is working or not

Diode Models (contd.)



Diode Models (contd.)

- Since, barrier potential and dynamic resistance is neglected, the voltage across diode in forward bias is zero and current can be calculated as

$$I_F = \frac{V_{Bias}}{R_{Limit}}$$

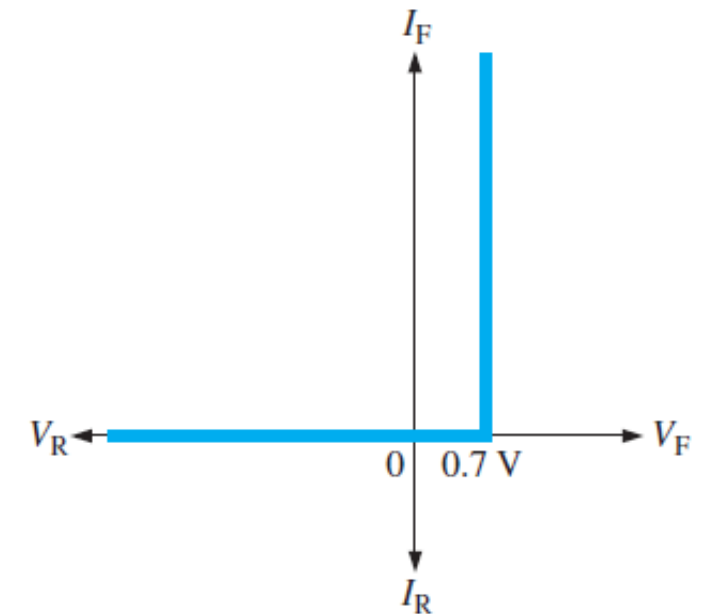
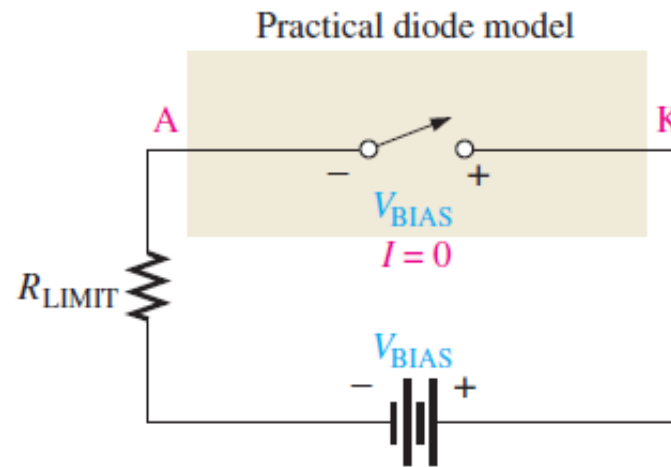
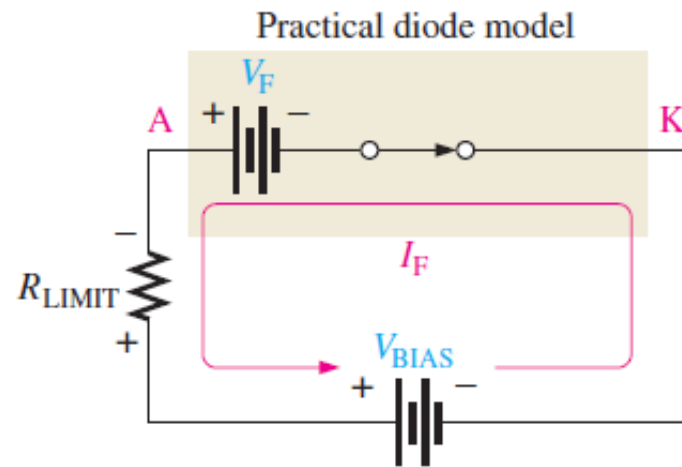
- Since, reverse current is neglected, means reverse current is zero and reverse voltage is equal to the bias voltage

$$I_R = 0 \quad \text{and} \quad V_R = V_{Bias}$$

Diode Models (contd.)

- **Practical Diode Model:**
 - In this approximation, the barrier potential is considered i.e. 0.7V for Si
 - In forward bias, a voltage source is considered with a closed switch
 - The +ive side of the source is at anode
 - Bias voltage should be greater than that voltage source in order to conduct a diode
 - In reverse bias, voltage source will not effect the circuit as diode acts as an open switch

Diode Models (contd.)



Diode Models (contd.)

- As diode has a voltage drop of 0.7V so

$$V_F = 0.7V$$

- The current through the diode can be calculated by KCL, hence

$$I_F = \frac{V_{Bias} - V_F}{R_{Limit}}$$

- In reverse bias, reverse current is zero and reverse voltage is equal to the bias voltage

$$I_R = 0 \quad \text{and} \quad V_R = V_{Bias}$$

- This approximation is useful when dealing with the low voltage calculations and designing basic diode circuits